



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis (MAT221)

Lebesgue's Criterion for Integrability

Measure Theory, Lebesgue's Theorem

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CONDUCTED BY

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Sets of Measure Zero

Theorem

A set $A \subseteq \mathbb{R}$ is said to be of measure zero if for every $\epsilon > 0$, there exists a countable collection of open intervals $\{U_n\}_{n=1}^{\infty}$ such that

$$A \subseteq \bigcup_{n=1}^{\infty} U_n \text{ and } \sum_{n=1}^{\infty} \mu(U_n) < \epsilon,$$

where $\mu(U_n)$ denotes the measure of the interval U_n .

💡 Countable Sets are of Measure Zero

Every countable set $A \subseteq \mathbb{R}$ is of measure zero.



Continuity and Oscillation

Oscillation of a Function

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function. The oscillation of f at a point $x_0 \in [a, b]$ is defined as

$$\omega_f(x_0) = \inf_{\delta > 0} \left(\sup_{|x - x_0| < \delta} f(x) - \inf_{|x - x_0| < \delta} f(x) \right).$$

Theorem

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function. Then f is continuous at a point $x_0 \in [a, b]$ if and only if the oscillation of f at x_0 is zero, i.e., $\omega_f(x_0) = 0$, where

$$\omega_f(x_0) = \inf_{\delta > 0} \left(\sup_{|x - x_0| < \delta} f(x) - \inf_{|x - x_0| < \delta} f(x) \right)$$

Lebesgue's Theorem

Theorem

A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable if and only if the set of points where f is discontinuous has measure zero.

Thank You!

We'd love your questions and feedback.

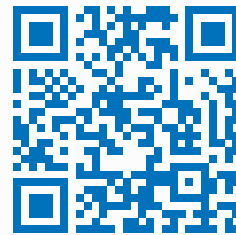
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(Lectures, walkthroughs, and course updates)



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References

- [1] Stephen Abbott, *Understanding Analysis*, 2nd Edition, Springer, 2015.
- [2] Terence Tao, *Analysis I*, 3rd Edition, Texts and Readings in Mathematics, Hindustan Book Agency, 2016.